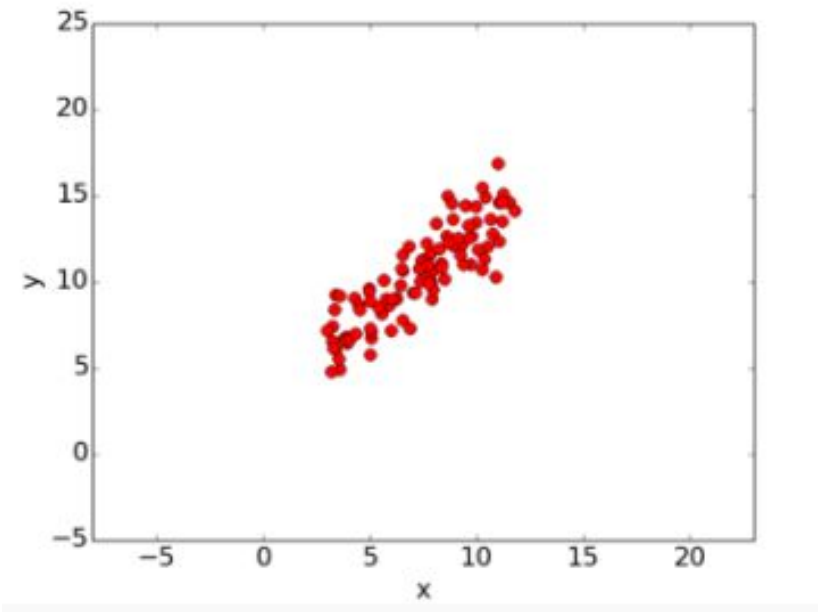
Regression Problem

- We will give an example for regression.
- We give closed form solution for regression.
- We give a numerical solution for regression.

Linear regression – an example that uses gradient descent



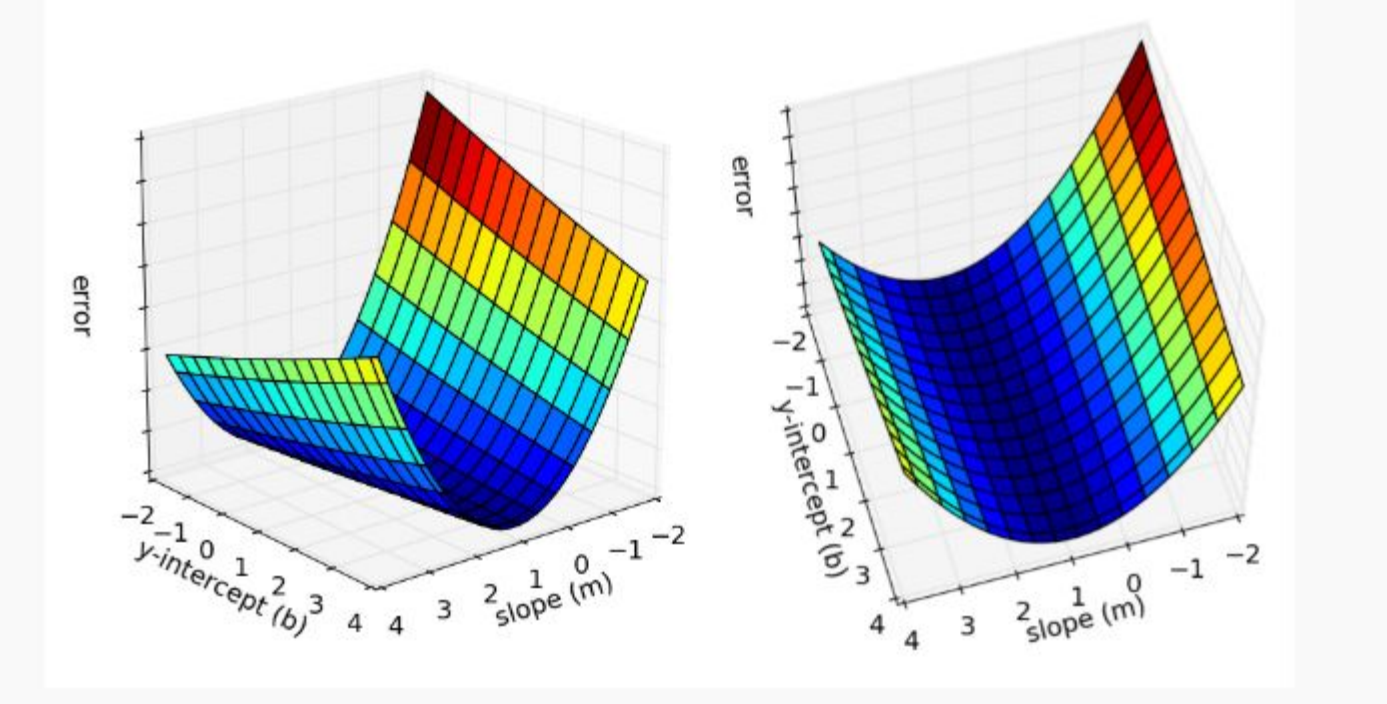
$$y = mx + b$$

We want to fit a straight line (in 2D case). The sample we are given with is $\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$.

We want to find (m, b).

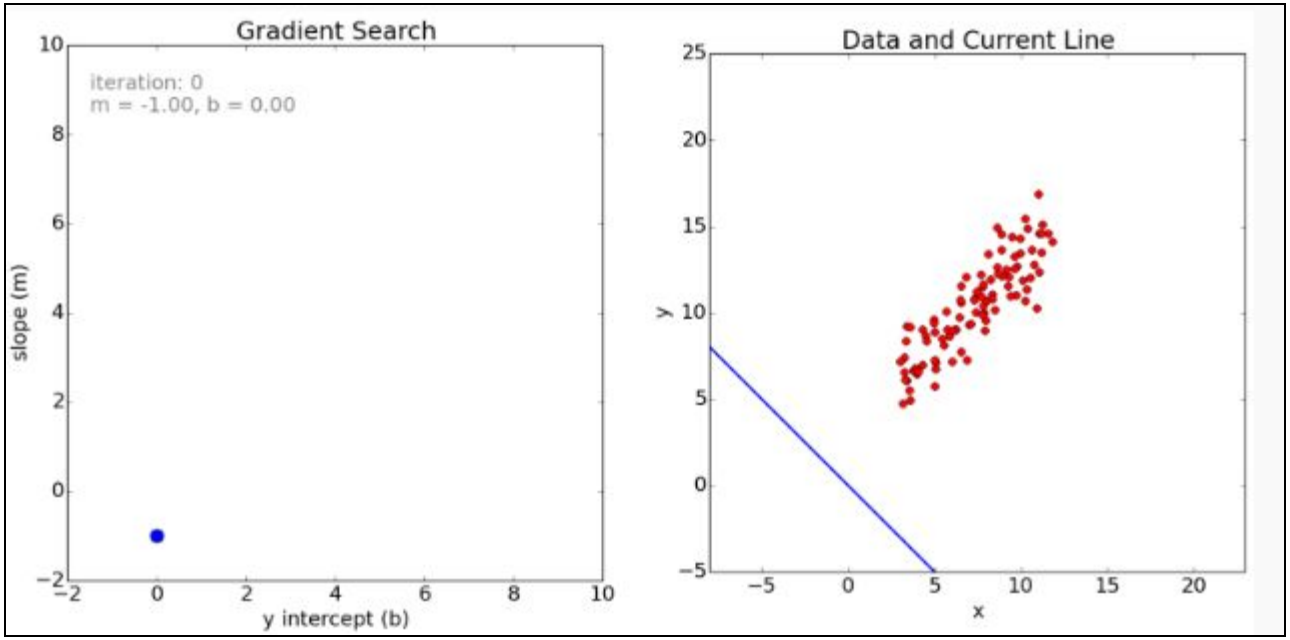
In the space (m, b) what is the function we are going to define? Minimum value of that function should be a solution for us.

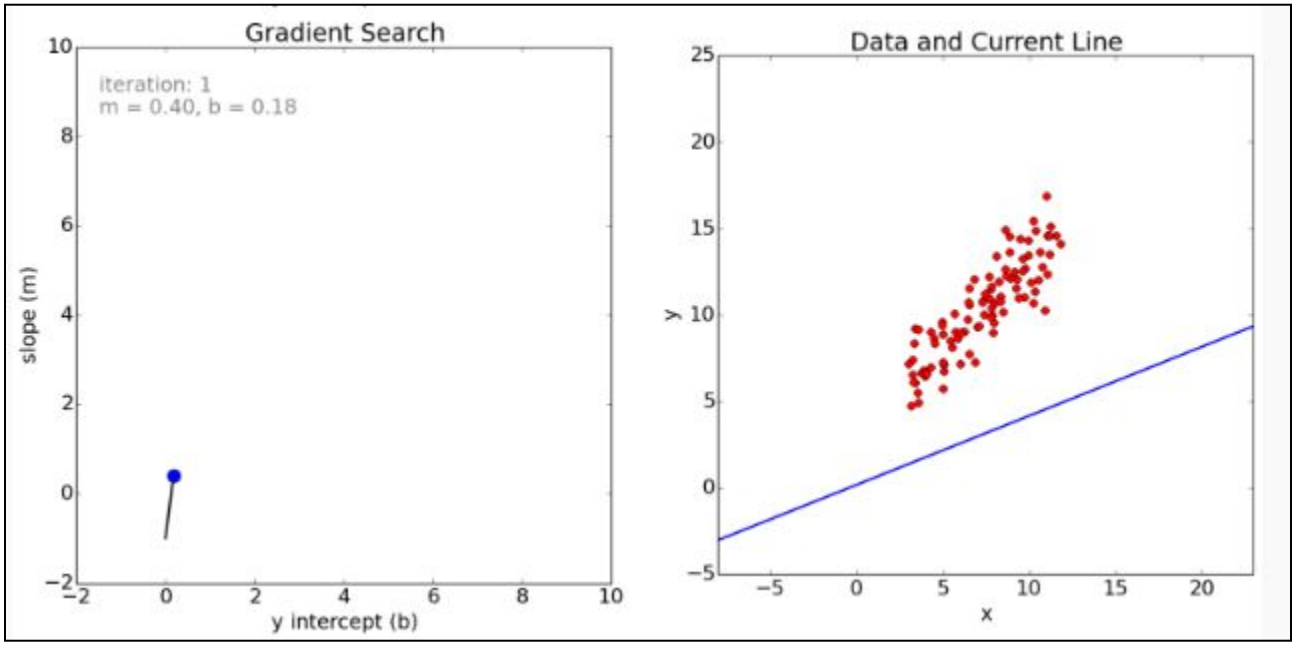
$$\text{Error}_{(m,b)} = \frac{1}{N}\sum_{i=1}^{N}(y_i - (mx_i + b))^2$$

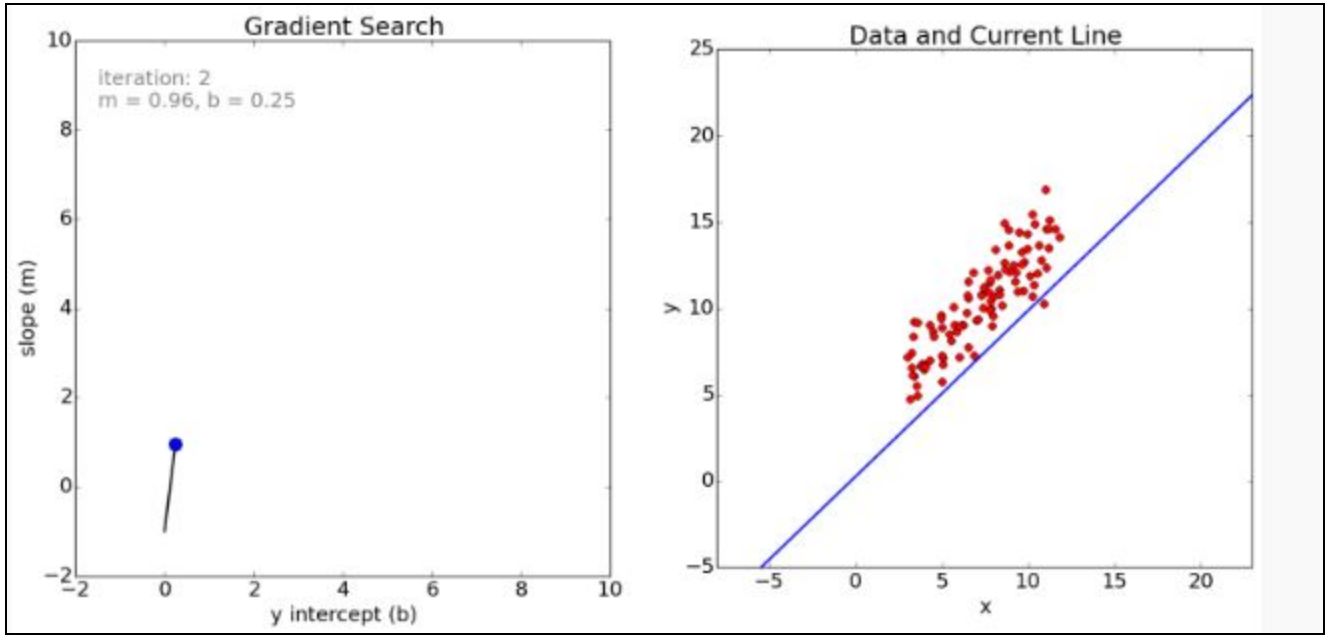


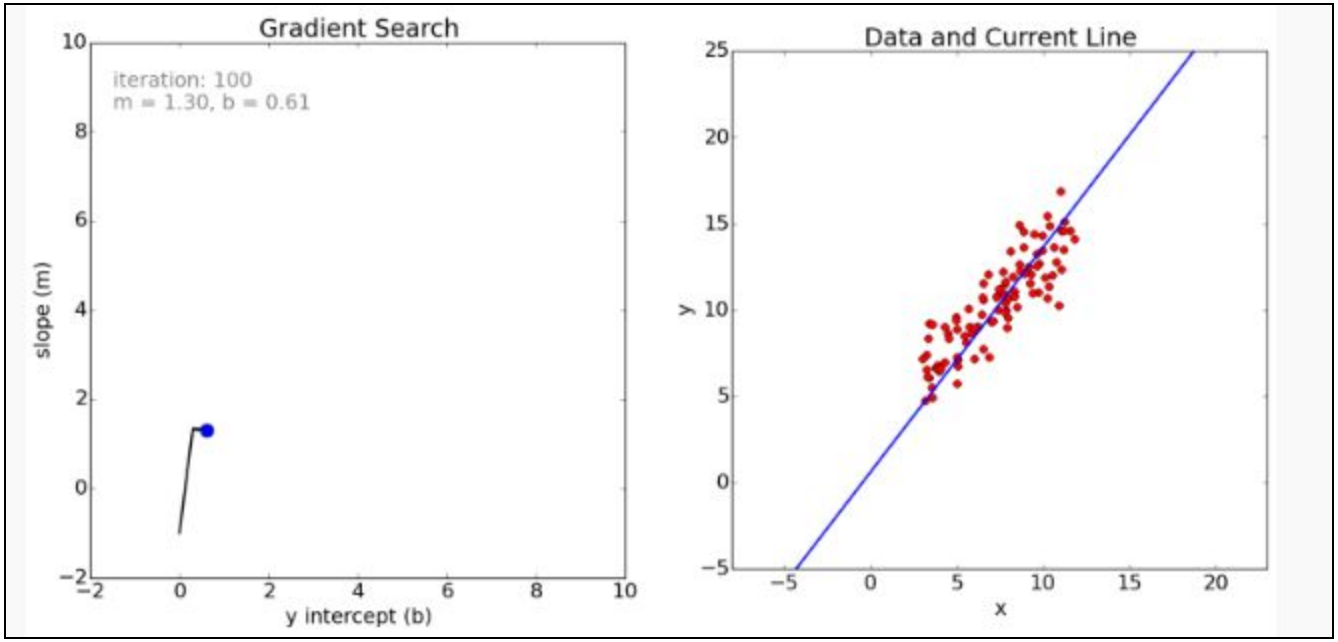
$$\frac{\partial}{\partial \mathrm{m}} = \frac{2}{N}\sum_{i=1}^{N} -x_i(y_i - (mx_i + b))$$

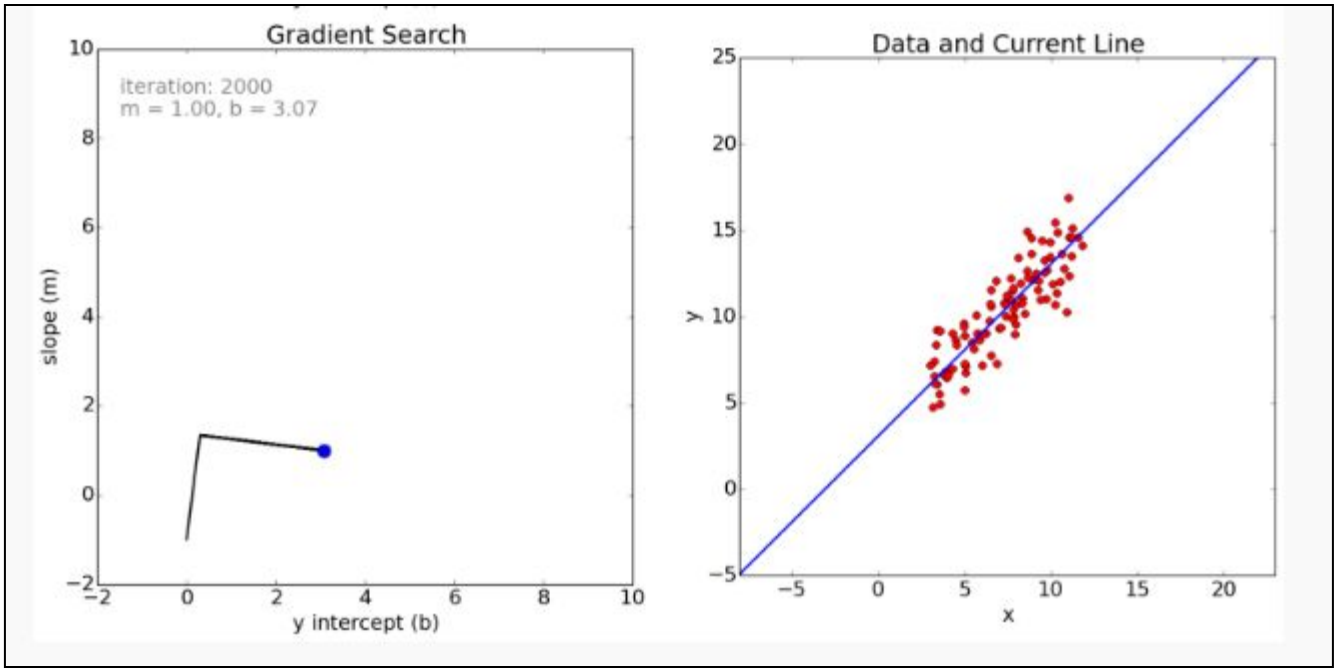
$$\frac{\partial}{\partial \mathrm{b}} = \frac{2}{N}\sum_{i=1}^{N} -(y_i - (mx_i + b))$$




Gradient Search
iteration: 1
m = 0.40, b = 0.18
slope (m)
y intercept (b)
Data and Current Line
y
x


Gradient Search
iteration: 2
m = 0.96, b = 0.25
slope (m)
y intercept (b)
Data and Current Line
y
x


Gradient Search
iteration: 100
m = 1.30, b = 0.61
slope (m)
y intercept (b)
Data and Current Line
y
x


Gradient Search
iteration: 2000
m = 1.00, b = 3.07
slope (m)
y intercept (b)
Data and Current Line
y
x

Closed form solution

- Since the criterion that is minimized is quadratic, the linear regression problem must be having a closed form solution.
- This is nothing but applying the Newton's descent method.

Linear Regression

Closed Form Solution

Notation

- Let the given data is $\mathrm{D} = \{(X_1, y_1), \ldots, (X_n, y_n)\}$.
- Let $X_i \in R^d$ and $y_i \in R$
- Further, $X_i = [x_{i1} \quad \ldots \quad x_{id}]^t$

We believe that

- $\theta_0 + \theta_1 x_{i1} + \cdots + \theta_d x_{id} = y_i$

We like to solve

- $$\begin{gathered}\theta_0 + \theta_1 x_{11} + \cdots + \theta_d x_{1d} = y_1 \\ \theta_0 + \theta_1 x_{21} + \cdots + \theta_d x_{2d} = y_2 \\ \vdots \\ \theta_0 + \theta_1 x_{n1} + \cdots + \theta_d x_{nd} = y_n\end{gathered}$$

In matrix form $Z\Theta = Y$

Solve this to find Θ

Z may not be invertible !!

$$Z\Theta = Y$$

- $$\begin{bmatrix} 1 & x_{11} & \dots & x_{1d} \\ 1 & x_{21} & \dots & x_{2d} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & \dots & x_{nd} \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Closest Solution: by minimizing the squared error

- Obtained by minimizing the criterion

$$J(\Theta) = \|Z\Theta - Y\|^2 = (Z\Theta - Y)^t(Z\Theta - Y)$$

$$J(\Theta) = \Theta^t Z^t Z\Theta - 2(Z\Theta)^t Y - Y^t Y$$

$$J(\Theta) = \Theta^t Z^t Z\Theta - 2(Z\Theta)^t Y - Y^t Y$$

- Equating gradient of J to zero. So

$$\nabla_\Theta(\,J\,) = 2(Z^t Z)\Theta - 2\mathrm{Z}^{\mathrm{t}}\mathrm{Y} = 0$$

We get $$\Theta = (Z^t Z)^{-1} Z^t Y$$

Example: 1D problem

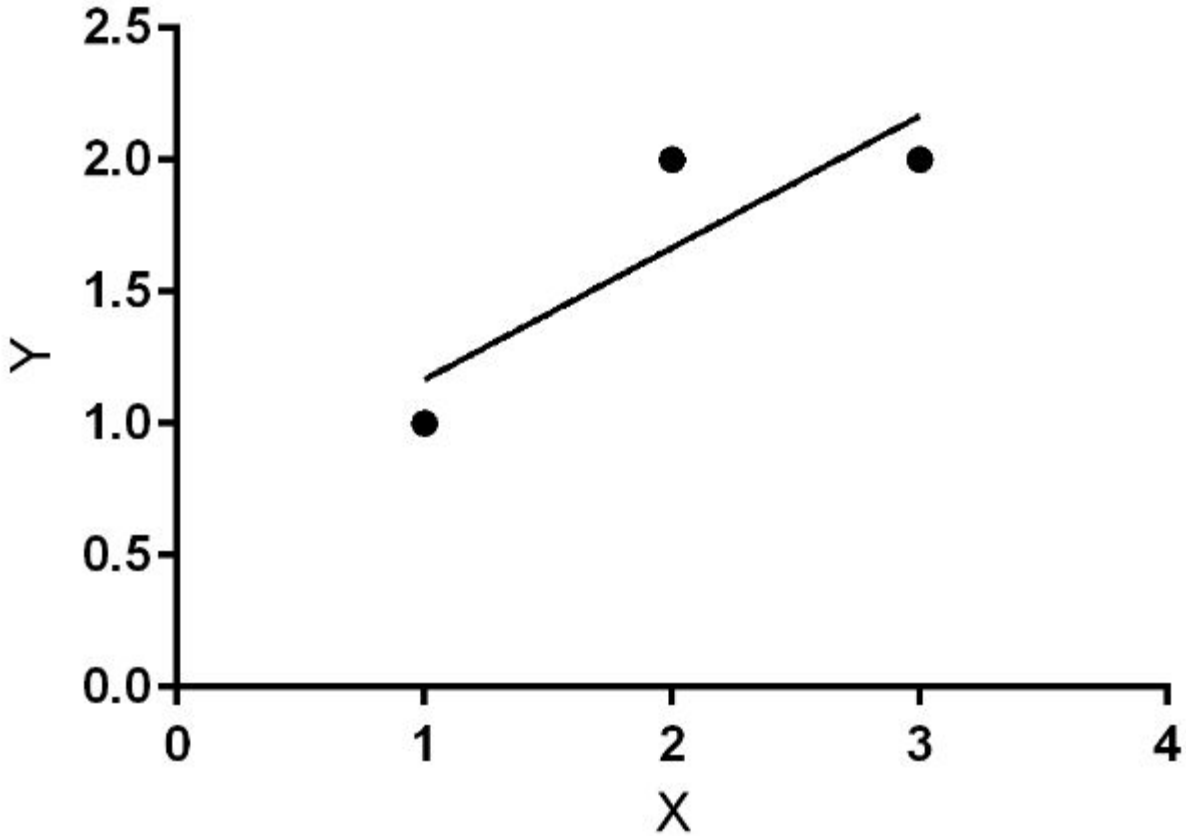
- $D = \{(1,1), (2,2), (3,2)\}$

- $Z = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \Theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}, Y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

- $Z^t Z = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \quad (Z^t Z)^{-1} = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$

- $\Theta = (Z^t Z)^{-1} Z^t Y$
- $\Theta = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}\begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/2 \end{bmatrix}$

- So the line we fitted is $y = \frac{2}{3} + \frac{x}{2}$


2.5
2.0
1.5
1.0
0.5
0.0
Y
0
1
2
3
4
X

Linear Regression – Numerical Solution

Iterative solution

Notation

- Let the given data is $\mathrm{D} = \{(X_1, y_1), \dots, (X_n, y_n)\}$.
- Let $X_i \in R^d$ and $y_i \in R$
- Further, $X_i = [x_{i1} \quad \dots \quad x_{id}]^t$
- We augment each X_i with 1 in order to simplify.
- The augmented vector we call Z_i

-
- $Z_i = [1 \;\; x_{i1} \;\; \cdots \;\; x_{id}]^t = \begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{id} \end{bmatrix}$

We like to solve

- $$\begin{gathered}\theta_0 + \theta_1 x_{11} + \cdots + \theta_d x_{1d} = y_1 \\ \theta_0 + \theta_1 x_{21} + \cdots + \theta_d x_{2d} = y_2 \\ \vdots \\ \theta_0 + \theta_1 x_{n1} + \cdots + \theta_d x_{nd} = y_n\end{gathered}$$

In matrix form $\boldsymbol{Z}\Theta = Y$

-

- $Z_i = \begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{id} \end{bmatrix} = [1 \quad x_{i1} \quad \dots \quad x_{id}]^t$

- $\boldsymbol{Z} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1d} \\ 1 & x_{21} & \dots & x_{2d} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & \dots & x_{nd} \end{bmatrix} = \begin{bmatrix} -Z_1 - \\ \vdots \\ -Z_n - \end{bmatrix} = \begin{bmatrix} Z_1^t \\ \vdots \\ Z_n^t \end{bmatrix}$

$$Z\Theta = Y$$

-
- $\boldsymbol{Z} = \begin{bmatrix} -Z_1 - \\ \vdots \\ -Z_n - \end{bmatrix} = \begin{bmatrix} Z_1^t \\ \vdots \\ Z_n^t \end{bmatrix}$
- $Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$
- $\Theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$

Criterion J

- $J(\Theta) = \|\boldsymbol{Z}\Theta - Y\|^2 = (\boldsymbol{Z}\Theta - Y)^t(\boldsymbol{Z}\Theta - Y)$

$$J(\Theta) = \Theta^t \boldsymbol{Z}^t \boldsymbol{Z}\Theta - 2(\boldsymbol{Z}\Theta)^t Y + Y^t Y$$

$$J(\Theta) = \Theta^t \boldsymbol{Z}^t \boldsymbol{Z}\Theta - 2(\boldsymbol{Z}\Theta)^t Y + Y^t Y$$

- $$\nabla_\Theta(J) = 2(\boldsymbol{Z}^t \boldsymbol{Z})\Theta - 2\boldsymbol{Z}^{\mathrm{t}} Y$$

Recall by equating the above to 0, we got the closed form solution which is,

$$\Theta = (\boldsymbol{Z}^t \boldsymbol{Z})^{-1} \boldsymbol{Z}^t Y$$

Iterative solution

- $\Theta_{k+1} = \Theta_k - \eta \nabla(J)$
- Note, this Gradient is at Θ_k

Batch Learning

- In each iteration entire training set is considered.
- $\Theta_{k+1} = \Theta_k - \eta(2(\boldsymbol{Z}^t\boldsymbol{Z})\Theta_k - 2\boldsymbol{Z}^tY)$
  $= \Theta_k - 2\eta\boldsymbol{Z}^t(\,\boldsymbol{Z}\Theta_k - Y\,)$

- Each element of Θ can be updated as below.
- $\theta_j = \theta_j - 2\eta\sum_{i=1}^{n}(h(Z_i) - y_i)x_j$

Stochastic Gradient Descent

- This is single sample correction method
- Consider one example, viz., (X_i, y_i)
- For X_i, its augmented vector is Z_i
- $Z_i = \begin{bmatrix} 1 \\ X_i \end{bmatrix}$
- $\Theta_{k+1} = \Theta_k - \eta\left(2\left(Z_i Z_i^t\right)\Theta_k - 2y_i Z_i\right)$

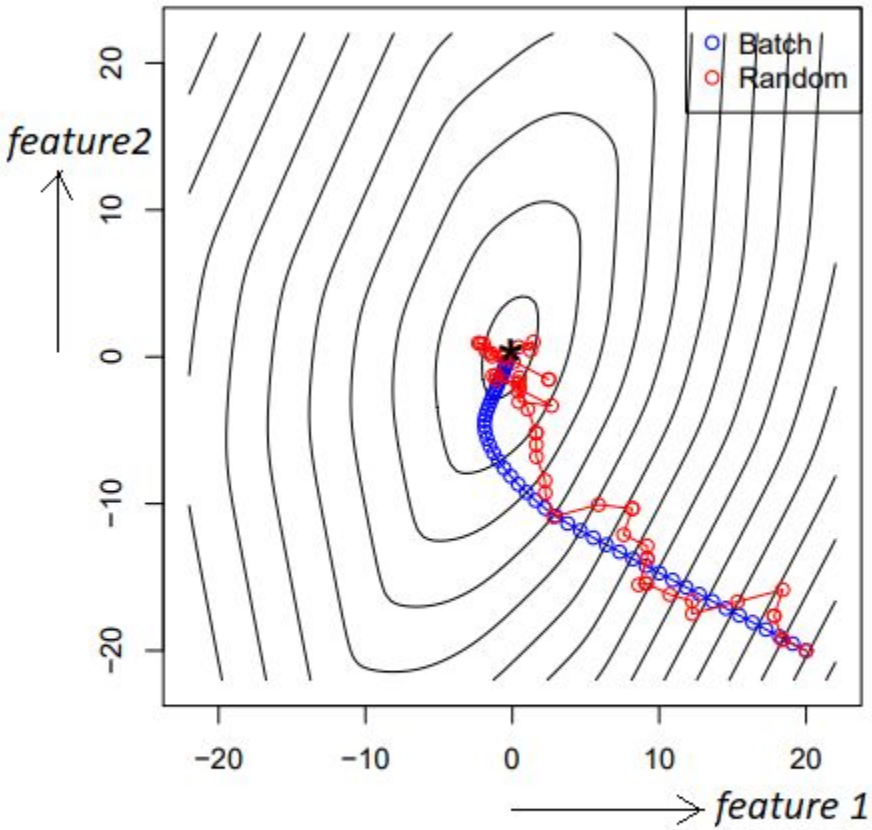
Normally X_i is chosen randomly from the training set.

Stochastic Gradient Descent

- $$\Theta_{k+1} = \Theta_k - \eta\left(2\left(Z_i Z_i^t\right)\Theta_k - 2y_i Z_i\right)$$
$$= \Theta_k - 2\eta\left(Z_i^t \Theta_k - \mathrm{y}_i\right)Z_i$$

- Each element of Θ can be updated as below.
- $\theta_j = \theta_j - 2\eta(h(Z_i) - y_i)x_j$

Batch Vs Stochastic Convergence



Blue: batch steps,
Red: stochastic steps,

Stochastic is also known as random / single sample correction.

- Batch method smoothly converges
- Stochastic method strays away often from the optimal path.

Stopping Condition

- It is better to have a validation set (VS) and
  - keep track of error on the VS,
  - stop when the error on the VS is not decreasing considerably
- In practice we fix number of iterations.

Polynomial Regression

- Same linear regression knowledge can be extended to achieve polynomial regression.
- Let the training data $= \mathrm{D} = \{ (1, 9), (0, 4), (2, 18)\}$
- Let us say that we want to fit a quadratic curve.
- The curve looks like $\theta_0 + \theta_1 x^2 + \theta_2 x$
- If the data points are (x^2, x), then the above formula is linear only.

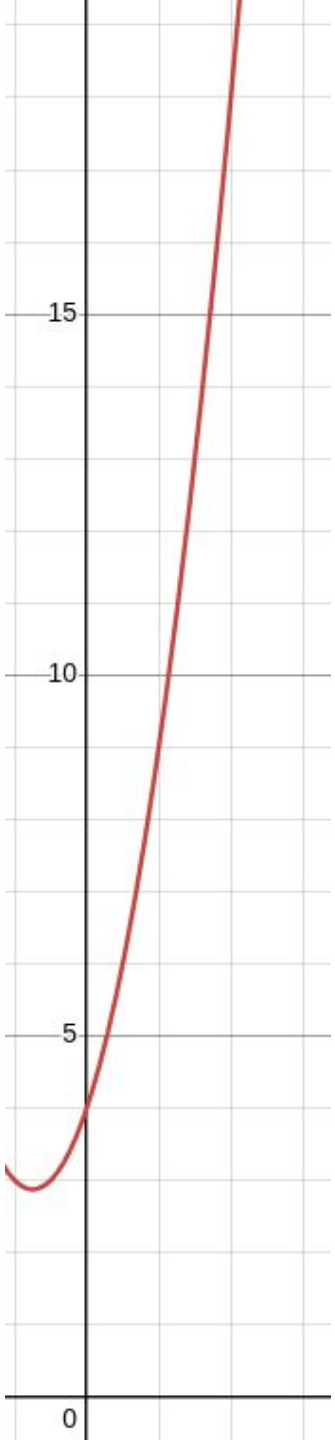
- $D = \{ (1,9), (0,4), (2,18) \}$
- The modified data would be

x^2	x	y
1	1	9
0	0	4
4	2	18

- We like to find $\theta_0 + \theta_1 x^2 + \theta_2 x = y$
- $Z = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 4 & 2 \end{bmatrix}, \; Y = \begin{bmatrix} 9 \\ 4 \\ 18 \end{bmatrix}$

-
- $Z^tZ = \begin{bmatrix} 3 & 5 & 3 \\ 5 & 17 & 9 \\ 3 & 9 & 5 \end{bmatrix}$
- $(Z^tZ)^{-1} = \begin{bmatrix} 1 & 1/2 & -3/2 \\ 1/2 & 3/2 & -3 \\ -3/2 & -3 & 13/2 \end{bmatrix}$

- $(Z^tZ)^{-1}Z^tY =$
$$\begin{bmatrix} 1 & 1/2 & -3/2 \\ 1/2 & 3/2 & -3 \\ -3/2 & -3 & 13/2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 9 \\ 4 \\ 18 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$$
- That is, $\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$
- The required function = $\theta_0 + \theta_1 x^2 + \theta_2 x$
$$= 4 + 2x^2 + 3x$$


15
10
5
0